

Braking Reaction Time After Right-Knee Anterior Cruciate Ligament Reconstruction: A Comparison of 3 Grafts

Bradley R. Wasserman, M.D., Brian C. Singh, B.S., Daniel J. Kaplan, B.A.,
Maxwell Weinberg, M.D., Robert Meislin, M.D., Laith M. Jazrawi, M.D., and
Eric J. Strauss, M.D.

Purpose: To determine when patients recover the ability to safely operate the brakes of an automobile after a right-knee anterior cruciate ligament reconstruction (ACLR). **Methods:** A computerized driving simulator was used to determine braking ability after an isolated right-knee ACLR. Thirty healthy volunteers were tested at 1 visit to determine normal mean values, and 27 treatment subjects were tested at 1 week, 3 weeks, and 6 weeks after ACLR. Nine study subjects were treated with a patella tendon (BPTB) autograft, 9 were treated with a hamstring (HS) autograft, and 9 were treated with a tibialis anterior (TA) allograft. The driving simulator collected data on brake reaction time (BRT), brake travel time (BTT), and total brake time (TBT) at each visit. **Results:** The control group generated a BRT of 725 milliseconds, BTT of 2.87 seconds, and TBT of 3.59 seconds. At week 1, all treatment patients had significant differences compared with controls for BRT, BTT, and TBT, except the BTT of the HS group. At week 3, all measures for the allograft group and the BRT for both autograft groups were no longer significantly different compared with controls, but significant differences were found for TBT in the HS and BPTB groups ($P = .03$, $P = .01$). At week 6, BRT, BTT, and TBT were no longer significantly different for either the HS group or BPTB group. **Conclusions:** Patients who underwent a right-knee ACLR with a TA allograft regained normal braking times by week 3 postoperatively. In contrast, those treated with a BPTB or HS autograft demonstrated significantly delayed braking times at 3 weeks but returned to normal braking ability by week 6. Those treated with an autograft had an earlier return of normalized BRT than BTT. **Level of Evidence:** Level III, case-control series.

Driving an automobile requires adequate lower-extremity strength, endurance, and coordination, as well as the ability to rapidly respond to unexpected stimuli. It is important for health care providers to be cognizant of injuries and surgical procedures that may impair a patient's driving ability.^{1,2} Anterior cruciate ligament reconstruction (ACLR) is a relatively common surgery for young patients of driving age, who frequently ask when it is safe to drive after reconstructive surgery. Recovery from ACLR partially

depends on the type of graft used, which may influence the postoperative strength of the involved extremity and play a role in the return to safe driving.³ Currently, however, there are no official guidelines available to assist clinicians or patients determine the optimal time to return to driving a vehicle after right-knee ACLR.

Graft selection is not only important for knee stability but also for the potential donor-site morbidity that may impact postoperative driving ability. Three commonly used grafts for ACLR are the bone–patellar tendon–bone (BPTB) autograft, the quadrupled hamstring (HS) autograft, and the tibialis anterior (TA) allograft.^{3–5} Previous studies have shown that the BPTB autograft has a greater incidence of postoperative anterior knee pain, patella fracture, and extension deficit, whereas harvest of the HS autograft may cause decreased knee flexion strength at deep flexion angles.^{6,7} Those who receive an allograft are not at risk for donor-site morbidity and benefit from shorter surgery times, improved cosmesis, and easier rehabilitation. The literature, however, has demonstrated

From the NYU Hospital for Joint Disease, New York, New York, U.S.A.

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Address correspondence to Daniel J. Kaplan, B.A., NYU Hospital for Joint Disease, 301 East 17th Street, New York, NY 10003, U.S.A. E-mail: danieljameskaplan@gmail.com

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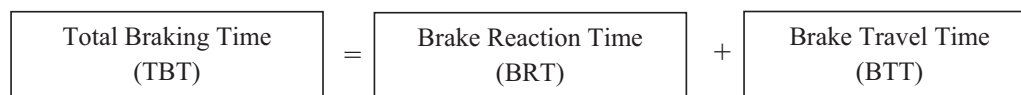


Fig 1. Components of TBT. BRT is the time it takes a driver to first touch the brake pedal after initially seeing a stop/emergency stimulus on the road. BTT is the interval between initial contact with the brake and the moment that the brake pedal reaches the end of its travel. These 2 parameters together equal the TBT time. BRT, brake reaction time; BTT, brake travel time; TBT, total brake time.

greater rates of re-rupture with allograft tissue.⁸ To date, 2 previous studies have evaluated driving ability after an ACLR with a validated driving simulator. One study examined only BPTB grafts after right-knee ACLR,⁹ whereas the second study evaluated driving after a right- or left-knee reconstruction with the use of mainly HS grafts.¹⁰

The purpose of the present study was to determine when patients recover the ability to safely operate the brakes of an automobile after a right-knee ACLR. We hypothesized that the brake reaction time (BRT: the time between the stimulus and initial contact with the brake), brake travel time (BTT: the interval between initial contact with the brake and the moment that the brake pedal reached the end of its travel), and total brake time (TBT: time between the initial stimulus and the moment the brake pedal reached the end of its travel) (Fig 1) would be delayed in postoperative ACLR patients compared with those with normal knees, and this difference would equalize by 6 weeks postoperatively. In addition, we hypothesized that patients who received an allograft reconstruction would have brake times equal to those of healthy control subjects earlier in the postoperative time period than patients who had autograft tissue harvested.

Methods

Study Population

In this Institutional Review Board–approved study, patients scheduled for a right-knee ACLR were identified from an institutional database. Patients older than 18 years of age with a valid driver's license and scheduled for a primary, isolated (i.e. the only knee pathology present was the anterior cruciate ligament [ACL] tear, and no meniscal or chondral procedures were performed), right-knee ACLR were included in the study. Patients were excluded if there was a history of previous knee surgery, if the time from injury to surgery was greater than 2 months, or if the patient had other medical problems that precluded safe and legal driving. Because the driving simulator used an automatic transmission that only required use of the right foot to brake, left-knee surgeries were excluded from enrollment.

Subjects were identified and recruited from the offices of 4 sports medicine fellowship–trained surgeons in the

morning before surgery with informed consent. Data on patients not attending all tests at any time during the testing period were excluded from the final analysis. Fifty-one patients were recruited initially for the study. Twenty-four patients were withdrawn from the final analysis because of missing data. Of the 27 subjects with complete data for analysis (15 female, 12 male), 9 underwent reconstruction with BPTB autograft, 9 underwent reconstruction with HS autograft, and 9 underwent reconstruction with TA allograft (Fig 2). ACL graft choice and fixation technique was surgeon-dependent and not influenced by participation in this study. BPTB and HS autografts underwent aperture fixation with screws on both sides. All allografts were fixed on the femur and tibia with an intrafix.

All treatment subjects underwent the same rehabilitation program. Immediately after surgery, patients were placed into a locked hinged knee brace and allowed to bear weight as tolerated on the operative extremity. At the initial postoperative visit (7-10 days after surgery), sutures were removed, the brace was unlocked, and patients were instructed to begin range of motion exercises both at home and under the



Fig 2. Flow diagram of subjects through the study.

guidance of a physical therapist. All patients discontinued the use of narcotics by the first driving trial. At 4 weeks postoperatively, when quadriceps control had been regained, the hinged knee brace was discontinued. All patients participated in proprioceptive and balance activities as part of the postoperative rehabilitation starting after the first postoperative visit. Proprioceptive training included various techniques, such as single-leg toe raises; single-leg balance with reaches; single-leg balance with eyes closed; single-leg leaning towers; or single-leg balance with various head, arm, trunk, and leg positions to challenge balance. Some variability in proprioceptive training was allowed on the basis of progress through the activities.

A group of 30 healthy volunteers (15 female, 15 male) with an asymptomatic right knee was recruited additionally to establish data in control subjects. Control subjects comprised volunteers at our institution who were not involved in data collection or in the health care delivery of study patients. Volunteers were recruited with use of an e-mail notification inviting them to participate and tested on the driving simulator anywhere from 4 weeks to 1 week before the treatment subjects. A group of 40 potential control subjects were invited to volunteer for the study, and 30 agreed to participate. Subjects were included in the control group if they were older than the age of 18 with a valid driver's license and were excluded if there was any history of right-knee surgery or other restrictions to driving a vehicle. The data gathered from the control subjects served as a representative population standard against which the data from the treatment subjects was compared. Healthy controls were established as the comparison group rather than paired preoperative times because preoperative study patients could potentially have inflated braking times as the result of their injury and inactivity.¹¹ The treatment subjects

underwent testing on a computerized driving simulator at 1 week (7-10 days), 3 weeks, and 6 weeks after ACL reconstruction. Controls were tested on the simulator in 1 sitting with 21 randomly assigned driving trials. Controls were only tested once because of the logistics of bringing in otherwise-healthy volunteers back to the simulator repeatedly.

Driving Simulation

The STISIM Drive (Systems Technology Inc. Simulator; Systems Technology, Inc., Hawthorne, CA) automotive simulation device is a fixed-base simulator equipped with an actual brake and accelerator pedal assembly attached to an automobile steering column (Fig 3A). An automobile seat adjustable for the patient's height was used. A 14-inch color monitor that displayed the driving simulation was placed in front of the steering wheel.

Fixed-base driving simulators such as the STISIM Drive have been used to establish the validity between on-road and simulated driver behavior in many key areas, including traffic rules, decision-making, acceleration, speed, and lateral control.^{12,13} STISIM Drive currently is used by various national and state agencies, including police academies and driving schools, as well as by several research companies in various settings.

All subjects sat in the simulator and adjusted the seat to a comfortable driving position. The computer displayed images of a speedometer and real-time relative values for the pedals (Fig 3B). Driving conditions (city, suburban, and highway) were simulated for each subject. Trials ranged in time from approximately 90 seconds to 5 minutes, and each subject performed a total of 21 randomly assigned trials at each visit. All participants were asked to drive the indicated speed limit, depending on the specific driving condition in each trial. Speed limits ranged from 15 miles per hour (mph) in the city

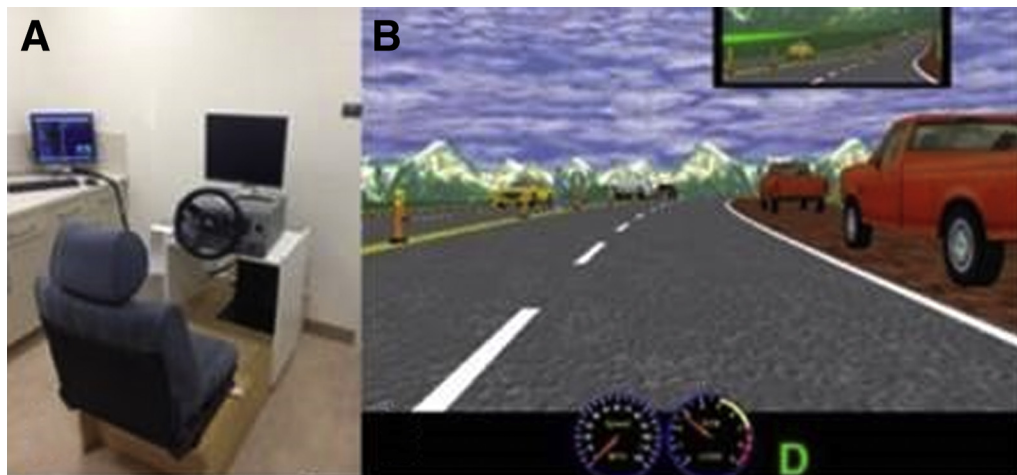


Fig 3. (A) Driving simulator hardware set-up. (B) Screenshot from Windows STISIM V2.0 driving simulator software, reproduced with permission.

Table 1. Demographics of the Controls and Treatment Groups

Demographics	Age, Years		Height, Inches		Weight, Pounds	
	Mean $\pm$ SD	<i>P</i> Value	Mean $\pm$ SD	<i>P</i> Value	Mean $\pm$ SD	<i>P</i> Value
Control (n = 30)	30.4 $\pm$ 3.8		68.6 $\pm$ 4.6		168 $\pm$ 31.7	
HS (n = 9)	29.2 $\pm$ 8.2	.970	66.2 $\pm$ 4.6	.481	158 $\pm$ 31.6	.817
BPTB (n = 9)	25.0 $\pm$ 4.2	.380	68.6 $\pm$ 2.3	1.000	167 $\pm$ 21.7	.999
TA (n = 9)	44.0 $\pm$ 11.4	.001	66.0 $\pm$ 3.3	.441	154 $\pm$ 32.4	.659

NOTE. *P* values were generated from pairwise comparison of each treatment subgroup with the control group.

BPTB, bone–patellar tendon–bone; HS, hamstring; TA, tibialis anterior.

setting to 55 mph in highway settings. At random times during each driving trial, a red stop sign appeared on the monitor to simulate an emergency stop situation. Participants were instructed to transfer the right foot from the accelerator pedal to the brake pedal as hard and as quickly as possible. Reaction times (BRT, BTT, TBT) were calculated by the driving software based on the individual's response to these random stop signs.

Statistical Analysis

The sample size in each treatment subgroup was determined through a power analysis that was calculated by the use of the standard deviation (622 milliseconds) and sample means (1,503–590 milliseconds) from the study of Nguyen et al.¹⁰ which evaluated BRT, and was similar to the current study. No similar studies in which the authors evaluated TBT could be found. Data from the right-sided ACLR study group was used (excluding the left-sided results) because this better matched our study population. Setting beta at 0.2 (80% power), we determined that a minimum of 7 participants were needed in each treatment subgroup (BPTB, HS, TA allograft).

Descriptive statistics, including the mean and standard deviation of the demographic data and braking times, were calculated. The TBT was the most clinically relevant outcome measure because it represented the time between initial stimulus and reaching a complete stop. The data were found to be not normally distributed and/or did not have homogeneously distributed variances as determined by the Shapiro-Wilk test ($P < .05$) and the Levene test for equality of variances ($P < .05$). Kruskal-Wallis tests were used to check for statistical differences between the 4 groups, as well as within each group along the various time points. When significant differences were found, Mann-Whitney *U* comparisons were used to directly compare subgroups. A *P* value of less than .05 was considered significant. All statistical tests were performed with SPSS, version 23 (IBM, Armonk, NY).

Results

Demographic data for the study and control subjects are shown in Table 1. There were no significant differences found between the 4 groups for patient

height and weight ($P = .311$, $P = .633$). The mean age of the TA group was statistically different than the control group ($P = .001$).

Mean data from the driving trials is presented in Table 2. At 7–10 days postoperatively, all treatment subgroups had significantly different (delayed) TBTs compared with the control group (Table 3). The BPTB group had the longest BTT and TBT but the shortest BRT of the treatment groups. Significant differences were found between all groups for BRT ($P = .001$), BTT ($P = .004$), and TBT ($P = .001$). On pairwise comparison, each treatment subgroup was statistically different from the controls for each outcome measure, except for BTT in the HS group ($P = .069$). Intergroup comparisons for the treatment subgroups showed no significant differences in any outcome measure.

At 3 weeks, statistically significant differences were found between groups for BTT ($P = .019$) and TBT ($P = .031$) but not BRT ($P = .622$), signifying normalized braking reaction among the treatment groups. Pairwise comparison of BTT and TBT data showed that the HS ($P = .025$, $P = .03$) and BPTB ($P = .008$, $P = .011$) subgroups remained significantly different than controls, whereas the TA subgroup had

Table 2. Means and Standard Deviations for Each Group at Week 1, Week 3, and Week 6

Graft	BRT, Seconds, Mean $\pm$ SD	BTT, Seconds, Mean $\pm$ SD	TBT, Seconds, Mean $\pm$ SD
Week 1			
Control (n = 30)	0.72 $\pm$ 0.09	2.87 $\pm$ 0.51	3.59 $\pm$ 0.56
HS (n = 9)	0.97 $\pm$ 0.22	3.41 $\pm$ 0.97	4.38 $\pm$ 1.09
BPTB (n = 9)	0.90 $\pm$ 0.19	3.97 $\pm$ 0.88	4.87 $\pm$ 0.96
TA (n = 9)	1.00 $\pm$ 0.24	3.44 $\pm$ 0.80	4.44 $\pm$ 0.95
Week 3			
Control (n = 30)	0.72 $\pm$ 0.09	2.87 $\pm$ 0.51	3.59 $\pm$ 0.56
HS (n = 9)	0.80 $\pm$ 0.16	3.42 $\pm$ 0.75	4.22 $\pm$ 0.88
BPTB (n = 9)	0.78 $\pm$ 0.15	3.42 $\pm$ 0.50	4.19 $\pm$ 0.61
TA (n = 9)	0.74 $\pm$ 0.14	3.04 $\pm$ 0.53	3.78 $\pm$ 0.64
Week 6			
Control (n = 30)	0.72 $\pm$ 0.09	2.87 $\pm$ 0.51	3.59 $\pm$ 0.56
HS (n = 9)	0.73 $\pm$ 0.09	3.11 $\pm$ 0.37	3.85 $\pm$ 0.38
BPTB (n = 9)	0.76 $\pm$ 0.19	3.38 $\pm$ 0.74	4.15 $\pm$ 0.87
TA (n = 9)	0.70 $\pm$ 0.14	2.93 $\pm$ 0.47	3.63 $\pm$ 0.58

BPTB, bone–patellar tendon–bone; BRT, brake reaction time; BTT, brake travel time; HS, hamstring; SD, standard deviation; TA, tibialis anterior; TBT, total brake time.

Table 3. Kruskal-Wallis and Mann-Whitney *U* Pairwise Comparisons With *P* Values for Each Group at Week 1, Week 3, and Week 6

Outcome Measure	<i>P</i> Value	BRT		<i>P</i> Value	BTT		<i>P</i> Value	TBT		<i>P</i> Value
Kruskal-Wallis Group Comparisons		Mann-Whitney <i>U</i> Pairwise Comparisons								
Week 1										
BRT	.001	Control <i>v</i>	HS	.001	Control <i>v</i>	HS	.069	Control <i>v</i>	HS	.023
BTT	.004		BPTB	.004		BPTB	.001		BPTB	.001
TBT	.001		TA	.001		TA	.042		TA	.004
Week 3										
BRT	0.622	Control <i>v</i>	HS	.299	Control <i>v</i>	HS	.025	Control <i>v</i>	HS	.030
BTT	0.019		BPTB	.315		BPTB	.008		BPTB	.011
TBT	0.031		TA	.658		TA	.501		TA	.588
Week 6										
BRT	0.973	Control <i>v</i>	HS	.857	Control <i>v</i>	HS	.114	Control <i>v</i>	HS	.107
BTT	0.167		BPTB	.806		BPTB	.080		BPTB	.064
TBT	0.146		TA	.781		TA	.857		TA	.961

NOTE. Bolded *P* values indicate statistical significance.

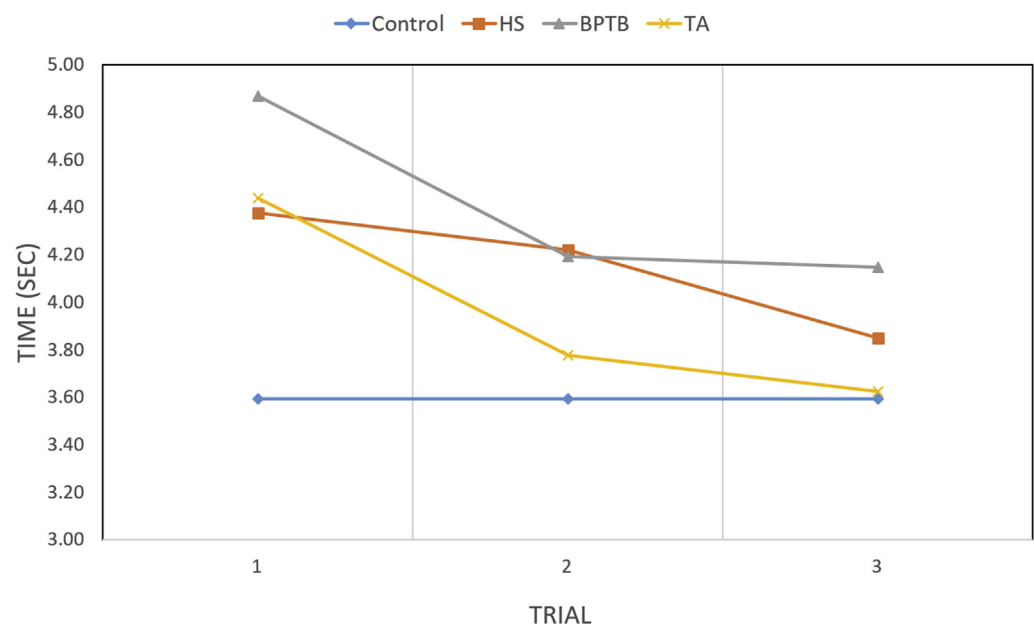
BPTB, bone–patellar tendon–bone; BRT, brake reaction time; BTT, brake travel time; HS, hamstring; TA, tibialis anterior; TBT, total brake time.

normalized times in every braking parameter. A statistically significant improvement from week 1 to week 3 was found for BRT in the TA subgroup ($P = .019$) but not for other subgroup data.

Intragroup data at 6 weeks showed improved braking times for each subgroup compared with 3 weeks, although none reached statistical significance. In addition, significant differences were no longer evident between treatment groups and controls in any outcome measure, indicating that total braking ability had equalized between each of the treatment and control groups. The BPTB group continued to have the longest BRT, BTT, and TBT. Intragroup comparisons through the course of the study (Fig 4) showed improved TBT for the TA group, HS group, and BPTB group.

Discussion

This study prospectively evaluated braking ability in patients after right-knee ACLR. On the basis of the results, we found that braking ability for allograft patients normalized at 3 weeks postoperatively compared with controls, as evidenced by significantly lower BRT, BTT, and TBT results. In the HS and BPTB groups, braking ability did not normalize until 6 weeks postoperatively when TBT was no longer significantly different than controls. In addition, data at 3 weeks showed that the HS and BPTB groups continued to have significant delays to the BTT but not the BRT, which suggests that autograft choice may have a greater impact on depressing the brake pedal than on BRT.

**Fig 4.** Measurements of total brake time by graft type.

The decision to allow a patient to return to driving after lower-extremity surgery is difficult—particularly with ACL patients, who are typically younger, when driving is often an integral part of their day-to-day lives. Physicians have an ethical and often legal responsibility to advise patients regarding a limitation that can potentially pose a danger to themselves or others.¹⁴ If a physician allows a patient to drive too early after surgery, it may put both the patient and the public at risk. The gravity of this decision demands that physicians use evidence-based literature when considering if a patient is ready to return to driving.

Previous studies from knee and hip replacement, or lower-extremity trauma, comprise the majority of evidenced-based literature regarding return-to-driving recommendations.^{1,15-20} There is a relative paucity of studies that consider driving after ACL reconstruction, particularly right-sided procedures. Gotlin et al.⁹ evaluated brake response time in patients who underwent ACLR with a patella tendon autograft but did not consider other graft types or the TBT. Nguyen et al.¹⁰ measured BRT in mainly HS grafts but also did not consider the TBT. It is difficult, however, to compare our data with previous autograft studies, given the variety of definitions for braking reaction time and the differing driving simulators. Nguyen et al.¹⁰ measured reaction time as the length of time between display of the red signal and the moment at which 200-Newton force was applied to the brake pedal, whereas Gotlin et al.⁹ defined reaction time as the transfer time from accelerator to the brake pedal. Nevertheless, both previous studies concluded that braking ability did not return to a level similar to controls until 6 weeks postoperatively. This finding is also in line with our conclusions for the patients in the autograft groups.

The TBT comprises both the BRT and BTT and is the most clinically relevant metric because it accounts for both reaction ability and the strength to depress the brake pedal. When TBT is translated into an increase in stopping distance compared with controls at 60 mph, the BPTB group took an additional 112 feet at week 1, 53 feet at week 3, and 48 feet at week 6. The HS group took an additional 69 feet at week 1, 55 feet at week 3, and 23 feet at week 6. The TA group took an additional 75 feet at week 1, 16 feet at week 3, and 3 feet at week 6. The use of healthy subjects as a control group is supported by the similarity between our control BRT (mean 720 milliseconds) and other preoperative and healthy groups reported in the literature with mean values of (736 milliseconds)¹¹ and (704 milliseconds).²¹ Although there is no universal standardized BRT for safe driving, the threshold for the United States was established by the Federal Highway Administration²² as 700 milliseconds, which is slightly lower than was found in our control group. Multiple factors, however, may influence BRT such as driver age, sex, level of

fatigue, cell phone use, task complexity, and environmental factors such as light level and traffic.

The BRT data at the first postoperative visit in the treatment subgroups are similar to data found by Hau et al.¹¹ at 1 week postoperatively in a study of braking reaction time from patients who underwent various right-knee arthroscopy procedures. As Hau et al.¹¹ did not include reconstructive procedures in their study, the similarity suggests that the recovery of BRT may be independent of the specific arthroscopic procedure performed. Data at 3 weeks in the current study showed that the HS and BPTB groups continued to have significantly different TBT from controls that was due to significantly different BTTs and not reaction times. The slower recovery in BTT for both of these treatment subgroups indicates that an autograft reconstruction may have a greater impact on the strength to completely depress the brake pedal but not the reaction time. Despite this finding, it is difficult to conclude whether this was attributable to donor-site morbidity or the reconstructive procedure itself. In addition, the earlier return to normal braking time in the TA group may have been influenced by the average older age in this group with more driving experience. One unexpected finding was the mean BTT in the HS group at week 1, which was not significantly different from controls.

Donor-site pain is associated with BPTB and HS autografts³ and may account for the significantly delayed improvement in braking times compared with the allograft treatment group. Although weakness from HS harvest and anterior knee pain after BPTB harvest may be seen at longer follow-up time periods,³ the initial postoperative pain can explain the significantly different TBT at earlier time points. Because the data showed an earlier return of BRT than BTT in the autograft groups, this indicates that transferring from the accelerator to the brake is a less-stressful activity for the operative extremity than flexing the foot against the brake pedal to completely stop the vehicle. Therefore, our data support the notion that allograft reconstructions result in faster recovery times (as it pertains to driving), and we attribute this to the lack of donor-site pain involved in the reconstruction. The earlier return to driving in the allograft group can be a consideration in some patients, in the context of the potential for greater rates of re-rupture.

Limitations

There are a few limitations of this study that bear mentioning. First is the relatively low sample size, which decreases the power of our study and increases the difficulty to detect differences. Although an a priori power analysis was performed, it could not be performed with the most clinically relevant metric of TBT, given the lack of available previous literature on this measurement.

Another limitation is the fact that the driving simulator was not an actual vehicle, and lack of patient factors accounted in the simulation testing. Driving speed, visual acuity, driving experience, patient age, and medications are additional variables that may affect braking times and subsequently safe driving. We found a significantly greater mean age in the TA group compared with other groups, which may influence the data as a result of differences in driving experience. Furthermore, physicians should remember that patients may drive a vehicle other than a car, such as public service vehicles and heavy goods vehicles, which demand more of the driver.^{14,23} Further potential selection bias may be present because ACL graft choice was surgeon-dependent. There also may be minor differences in surgical technique between surgeons. The last limitation is the possibility of a learning curve associated with the simulation. Although a learning curve was not considered in our analysis, it is possible that some of the improvement seen in successive driving trials may represent a cognitive phenomenon, which cannot be distinguished from postoperative recovery. This may be particularly present with BRT data, which is more prone to a learning curve aberration than BTT because an appropriate BRT requires a response to a unique stimulus. In addition, BTT was the main determinant of TBT (as BRT comprises a small percentage of total TBT), and this would be considerably less affected by a learning curve. Both the treatment and healthy control groups may have been subject to this learning curve, and it is possible mean control times at each week would decrease. Steps were taken, however, to minimize the learning curve phenomenon, such as multiple driving tracks and a randomly assigned order of driving trials.

Conclusions

Patients who underwent a right-knee ACLR with a TA allograft regained normal braking times by week 3 postoperatively. In contrast, those treated with a BPTB or HS autograft demonstrated significantly delayed braking times at 3 weeks but returned to normal braking ability by week 6. Those treated with an autograft had an earlier return of normalized BRT than BTT.

References

- Marecek GS, Schafer MF. Driving after orthopaedic surgery. *J Am Acad Orthop Surg* 2013;21:696-706.
- Goodwin D, Baecher N, Pitta M, Letzelter J, Marcel J, Argintar E. Driving after orthopedic surgery. *Orthopedics* 2013;36:469-474.
- Shelton WR, Fagan BC. Autografts commonly used in anterior cruciate ligament reconstruction. *J Am Acad Orthop Surg* 2011;19:259-264.
- Romanini E, D'Angelo F, De Masi S, et al. Graft selection in arthroscopic anterior cruciate ligament reconstruction. *J Orthop Traumatol* 2010;11:211-219.
- Shybut TB, Pahk B, Hall G, et al. Functional outcomes of anterior cruciate ligament reconstruction with tibialis anterior allograft. *Bull Hosp Jt Dis* (2013) 2013;71:138-143.
- Gifstad T, Foss OA, Engebretsen L, et al. Lower risk of revision with patellar tendon autografts compared with hamstring autografts: A registry study based on 45,998 primary ACL reconstructions in Scandinavia. *Am J Sports Med* 2014;42:2319-2328.
- McRae S, Leiter J, McCormack R, Old J, MacDonald P. Ipsilateral versus contralateral hamstring grafts in anterior cruciate ligament reconstruction: A prospective randomized trial. *Am J Sports Med* 2013;41:2492-2499.
- Kaeding CC, Pedroza AD, Reinke EK, Huston LJ, Consortium M, Spindler KP. Risk factors and predictors of subsequent acl injury in either knee after ACL reconstruction: Prospective analysis of 2488 primary ACL reconstructions from the MOON cohort. *Am J Sports Med* 2015;43:1583-1590.
- Gotlin RS, Sherman AL, Sierra N, Kelly M, Scott WN. Measurement of brake response time after right anterior cruciate ligament reconstruction. *Arthroscopy* 2000;16:151-155.
- Nguyen T, Hau R, Bartlett J. Driving reaction time before and after anterior cruciate ligament reconstruction. *Knee Surg Sports Traumatol Arthrosc* 2000;8:226-230.
- Hau R, Csongvay S, Bartlett J. Driving reaction time after right knee arthroscopy. *Knee Surg Sports Traumatol Arthrosc* 2000;8:89-92.
- Helman S, Reed N. Validation of the driver behaviour questionnaire using behavioural data from an instrumented vehicle and high-fidelity driving simulator. *Accid Anal Prev* 2015;75:245-251.
- Gibbons C, Mullen N, Weaver B, Reguly P, Bedard M. One- and three-screen driving simulator approaches to evaluate driving capacity: Evidence of congruence and participants' endorsement. *Am J Occup Ther* 2014;68:344-352.
- Nunez VA, Giddins GE. 'Doctor, when can I drive?': An update on the medico-legal aspects of driving following an injury or operation. *Injury* 2004;35:888-890.
- Dalury DF, Tucker KK, Kelley TC. When can I drive? Brake response times after contemporary total knee arthroplasty. *Clin Orthop Relat Res* 2011;469:82-86.
- Jordan M, Hofmann UK, Rondak I, Gotze M, Kluba T, Ipach I. Brake response time is significantly impaired after total knee arthroplasty: Investigation of performing an emergency stop while driving a car. *Am J Phys Med Rehabil* 2015;94:665-676.
- Abbas G, Waheed A. Resumption of car driving after total hip replacement. *J Orthop Surg (Hong Kong)* 2011;19:54-56.
- Jeng CL, Lin JS, Amoyal K, Campbell J, Myerson MS. Driving brake reaction time following right ankle arthrodesis. *Foot Ankle Int* 2011;32:896-899.
- Egol KA, Sheikhaazadeh A, Koval KJ. Braking function after complex lower extremity trauma. *J Trauma* 2008;65:1435-1438.

20. Jordan M, Hofmann UK, Grunwald J, et al. Influence of left- and right-side total hip arthroplasty on the ability to perform an emergency stop while driving a car. *Arch Phys Med Rehabil* 2014;95:1702-1709.
21. MacDonald W, Owen JW. The effect of total hip replacement on driving reactions. *J Bone Joint Surg Br* 1988;70:202-205.
22. Green M. How long does it take to stop? Methodological analysis of driver perception-brake times. *Transportation Human Factors* 2000;2:195-216.
23. Giddins GE, Hammerton A. "Doctor, when can I drive?" A medical and legal view of the implications of advice on driving after injury or operation. *Injury* 1996;27:495-497.